

UNIVERSITY OF BRISTOL
Summer Reassessment for Fall 2024
Topics in CS Question

SEMT30003/4
Computational Neuroscience

TIME ALLOWED:
1 hour

This paper contains one in-depth question with three sections that contain multiple parts. **All** components must be attempted and will be used for assessment. The maximum for this section is 50 marks.

Word counts are not mandatory, but rather hints to help you avoid spending more time than needed. You may exceed the advised word count if you wish. You may also use diagrams and mathematics as appropriate in short answer questions.

Please write your student number on the line below.
(To ensure blinded marking, do not write your name)

Other instructions:

- Only approved non-programmable calculators are permitted
- Write your answers in the question paper
- Blank pages are provided at the end if you need extra space. Clearly label which answer you are writing.

(a) [6 marks] General knowledge: (3–9 minutes, 1–3 minutes per question)

- i. Discuss rate and spike-timing codes in the context of calculation of inter-aural time delays by the brainstem, and in continuous modulation and control of muscle dynamics by motor neurons. Which scenario is better captured by rate coding? What are the key differences between them? (≈ 50 –100 words) **[2]**
- ii. Hubel and Wiesel's recordings from primary visual cortex (V1, Brodmann area 17) found visually tuned neurons they termed '*simple cells*'. What types of visual input do simple cell's respond to, and how do they differ from the so-called '*complex cells*'? (≈ 50 –100 words) **[2]**
- iii. Most retinal ganglion cells do not transmit absolute luminance or colour information. Instead, they spike less often and only when they detect changes over time, space, or a mixture of the two. Broadly, what is the name for this type of encoding and what are its advantages? (≈ 50 -100 words) **[2]**

(b) [22 marks] Leaky integrate-and-fire dynamics: (10–20 minutes)

Consider a Leaky Integrate-and-Fire (LIF) neuron obeying the following dynamics:

$$RC\dot{v} = E_\ell - v + IR$$

(1)

if $v(t) > v_{\text{threshold}}$ then $v(t) \leftarrow v_{\text{reset}}$ and emit a spike

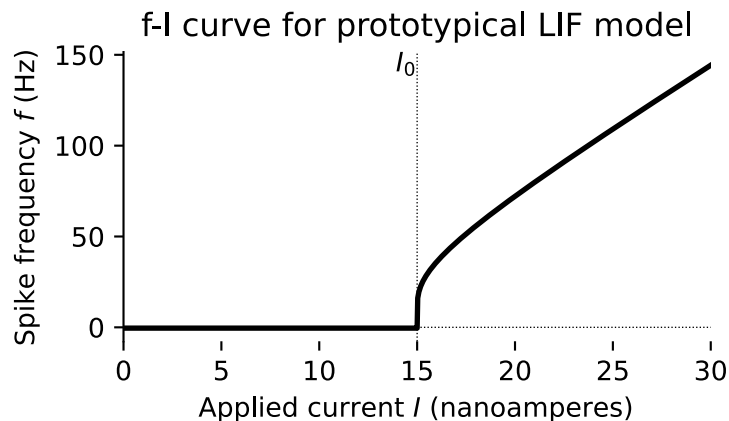
Assume that $v_{\text{reset}} \leq E_\ell < v_{\text{threshold}}$.

- i. Give the physiological interpretation of C , R , E_ℓ , I , $v_{\text{threshold}}$, and v_{reset} . Provide valid scientific units for each quantity, and discuss how this model approximates the ionic currents across the neuronal membrane during the action potential. (address all 5 variables ≈ 50 – 200 words) **[6]**
- ii. Show that Equation (2) (below) is the solution to the ODE in Equation (1) if the spiking threshold and reset are ignored:

$$v(t) = e^{-t/\tau}(v(0) - v_\infty) + v_\infty. \quad (2)$$

and obtain the values of τ and v_∞ in terms of C , R , E_ℓ , and/or I . You may use any appropriate method (verifying by differentiating, method of undetermined coefficients, integrating factor, or methods not covered in this course such as Fourier or Laplace transforms). **[6]**

- iii.α Below is a plot of firing frequency as a function of injected current. The LIF model has a hard threshold: The model abruptly starts spiking above a certain minimum current (labelled I_0 below). Derive an expression for I_0 in terms of v_∞ , C , R . **[5]**



Note: You do not need to know the values of the model parameters here. No calculation is required and you may leave your answer in terms of v_∞ , C , R .

- iii.β State whether you'd expect I_0 to increase or decrease if E_ℓ is *decreased*. Would this make it easier or harder for incoming excitatory synapses to cause this neuron to fire? Briefly speculate about what such a change might correspond to in a binary perceptron, whose output is given by $y = \Theta(\mathbf{w}^\top \mathbf{x} - b)$. (≈ 10 – 20 words) **[5]**

(c) [22 marks] The Hodgkin–Huxley Model: (10–20 minutes)

The voltage in the Hodgkin Huxley model of the action potential evolves according to the following differential equations:

$$\begin{aligned}
 C\dot{v} &= g_{\ell}(E_{\ell} - v) + \bar{g}_K n^4(E_K - v) + \bar{g}_{Na} m^3 h(E_{Na} - v) + I \\
 \dot{n} &= \alpha_n(v) \cdot (1 - n) - \beta_n(v) \cdot n \\
 \dot{m} &= \alpha_m(v) \cdot (1 - m) - \beta_m(v) \cdot m \\
 \dot{h} &= \alpha_h(v) \cdot (1 - h) - \beta_h(v) \cdot h.
 \end{aligned} \tag{3}$$

You do not need to know the specific form of the nonlinear functions $\alpha_{\dots}(v)$ and $\beta_{\dots}(v)$ for this question.

- i. What is the physiological interpretation of the constants $\bar{g}_{\ell}$, $\bar{g}_K$, $\bar{g}_{Na}$? Provide physical units for these variables and discuss their role in the voltage dynamics. (≈ 25 –50 words) **[2]**
- ii. What part of the action potential does the gating variable h on the sodium conductance capture? (≈ 25 –50 words) **[2]**
- iii. **Voltage clamp:** Recall that the functions $\alpha_{\dots}(v)$ and $\beta_{\dots}(v)$ can be experimentally measured by using a feedback amplifier to inject current into the neuron to hold $v(t)$ at a constant value, thereby setting $\alpha_{\dots}$ and $\beta_{\dots}$ to constant values, and turning the ODEs for gating variables n , m , h into stable first-order linear differential equations. This is called a “voltage clamp” recording. Throughout this sub-question, assume that voltage v and gating variables $\alpha_{\dots}$ and $\beta_{\dots}$ are unknown constants, and that the linear ODEs for the gating variables are stable.
 - iii.a For a given constant voltage v , derive an expression for the steady-state value for the gating variable h , denoted as $h_{\infty} = \lim_{t \rightarrow \infty} h(t)$ in terms of $\alpha_h(v)$ and $\beta_h(v)$. **[6]**
 - iii.b Re-write the differential equation for the sodium gating variable in the form

$$\tau_h(v)\dot{h} = h_{\infty}(v) - h \tag{4}$$

and give an expression for $\tau_h(v)$ in terms of $\alpha_h(v)$ and $\beta_h(v)$. **[6]**

- iii.c Pretend that you have measured $\tau_h(v)$ and $h_{\infty}(v)$ from a voltage-clamp recording at voltage v . Derive expressions to calculate $\alpha_h(v)$ and $\beta_h(v)$ from $\tau_h(v)$ and $h_{\infty}(v)$. **[6]**